

ECON 311

Lecture 19: AS-AD Model

Introduction

- This is the central macroeconomic model for understanding business cycles, which combines
 1. Firms' price setting behaviour
 2. Demand behaviour
 3. Government (CB) policy responses to economic shocks
- In the previous chapter we studied how government stabilization policy works to offset short-run deviations from potential output.
 - ▶ In this chapter we take stabilization policy as given and we study how the interaction of 1,2,3 results in short-run fluctuations of output and inflation.

AS-AD Model

- The IS-MP Model consists of the following three equations:

$$\text{IS curve:} \quad \tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1$$

$$\text{MP curve:} \quad \text{CB chooses: } R_t$$

$$\text{Phillips curve:} \quad \Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}_t$$

- As we previously studied, Central Banks react, by changing the interest rates (demand management), to deviations of inflation from a specific target:

$$\text{Policy rule:} \quad R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

- $\bar{m}$ denotes how aggressively monetary policy responds to inflation.

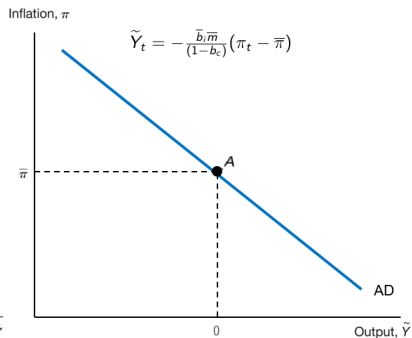
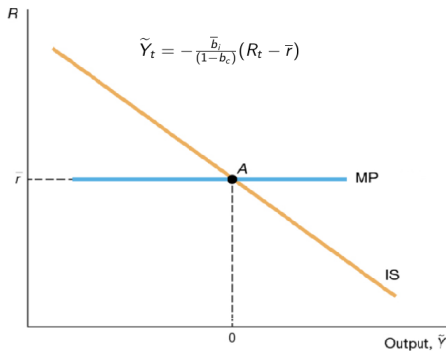
AD Curve

- We combine the policy rule with the IS curve to obtain:

$$\text{AD Curve: } \tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i \bar{m}(\pi_t - \bar{\pi})] - 1$$

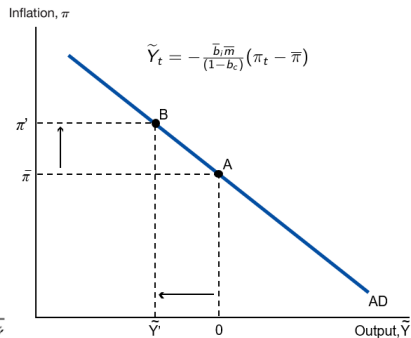
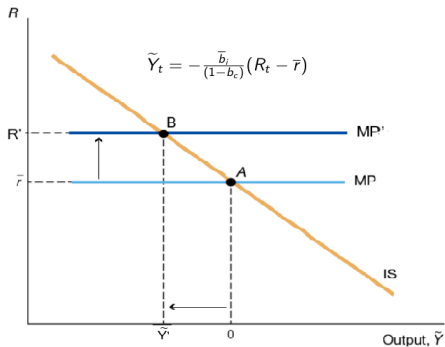
AD Curve

Benchmark IS, AD curves (i.e. $\bar{a} = 1 - b_c$)



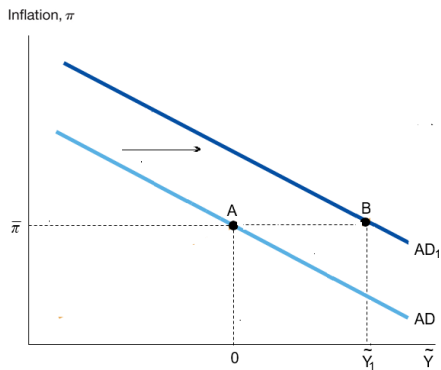
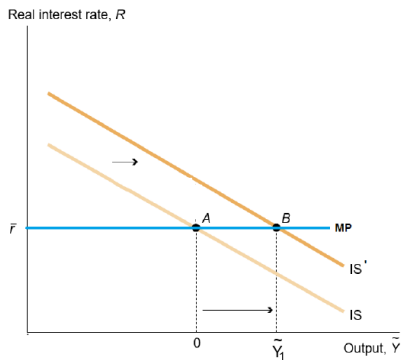
AD Curve

Benchmark IS, AD curves (i.e. $\bar{a} = 1 - b_c$)



AD Curve

IS, AD curves ($\bar{a} = 1 - b_c$, $\bar{a}_1 > \bar{a}$)



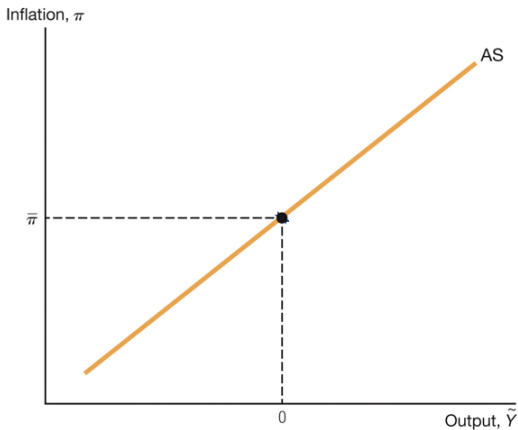
AS Curve

- We rewrite price setting behaviour from Phillips Curve as:

$$\text{AS Curve: } \pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t + \bar{o}_t$$

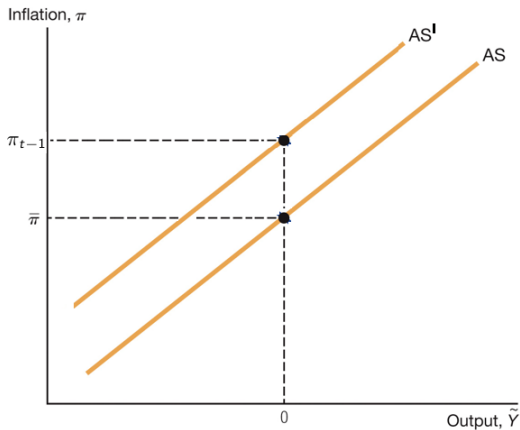
AS Curve

Benchmark AS curve (i.e. $\pi_{t-1} = \bar{\pi}$, $\bar{o}_t = 0$)



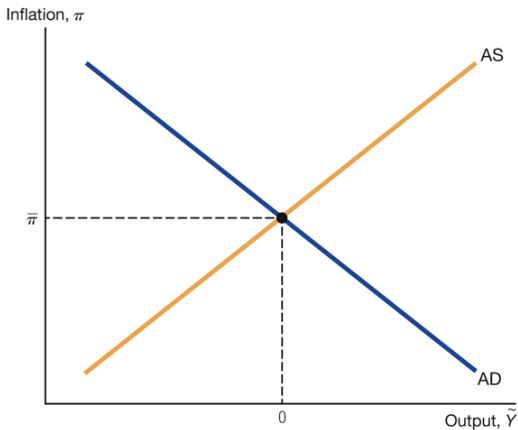
AS Curve

Benchmark AS curve (i.e. $\pi_{t-1} > \bar{\pi}$, $\bar{o}_t = 0$)



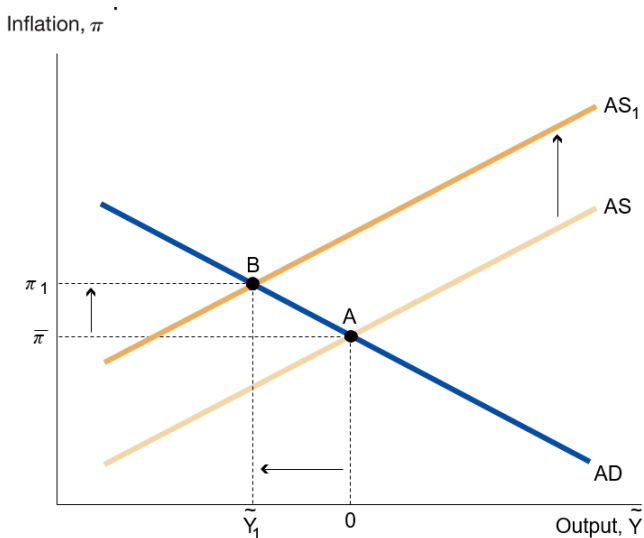
AS-AD Model

Benchmark AS curve ($\pi_{t-1} = \bar{\pi}$, $\bar{o}_t = 0$), AD ($\bar{a} = 1 - b_c$)

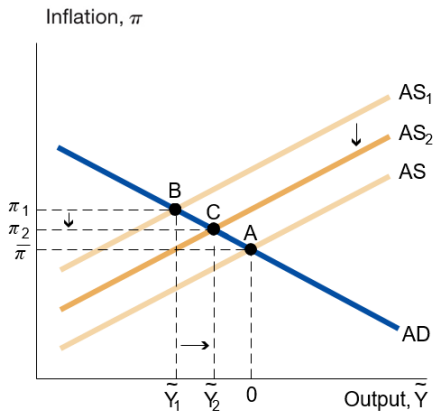
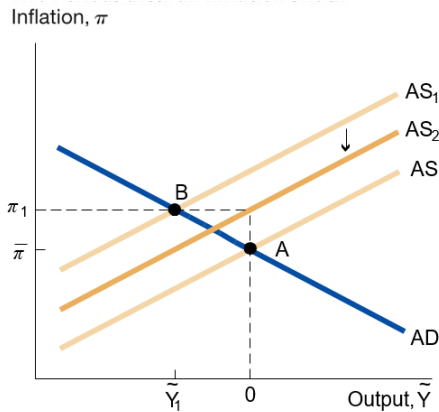


Using the AS-AD Model

Cost-push Shock

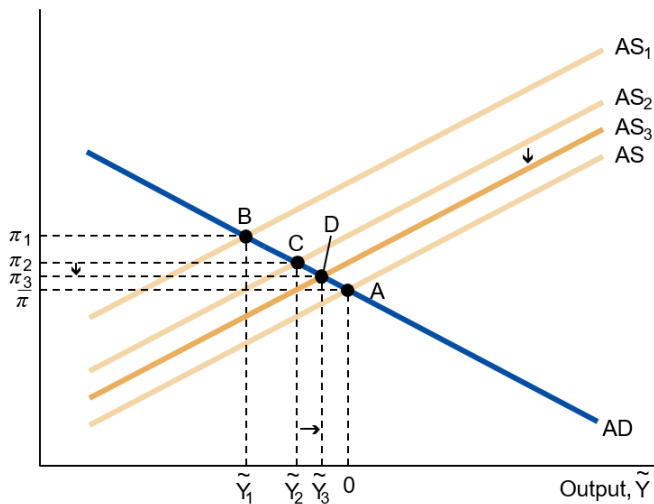


Inflation Shock

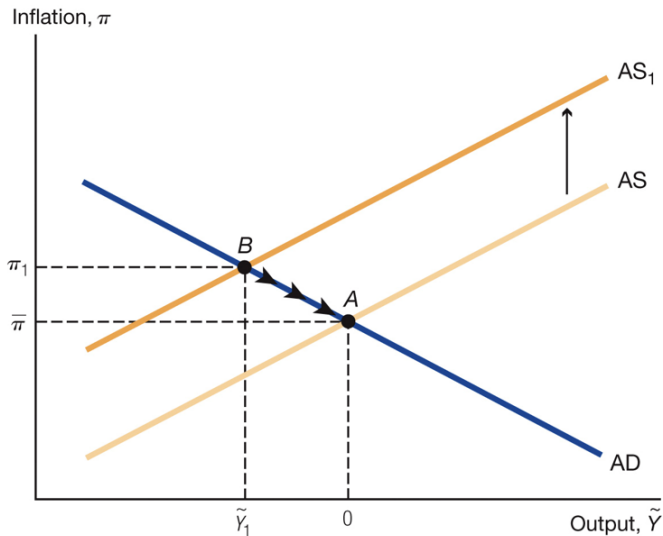


Cost-push Shock

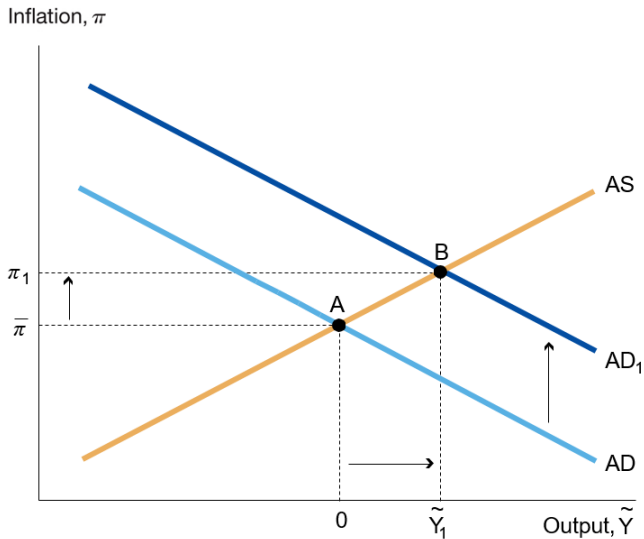
Inflation, π



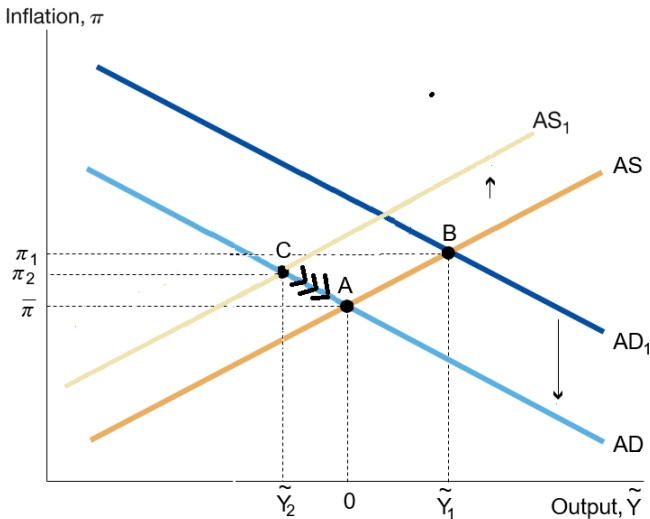
Cost-push Shock



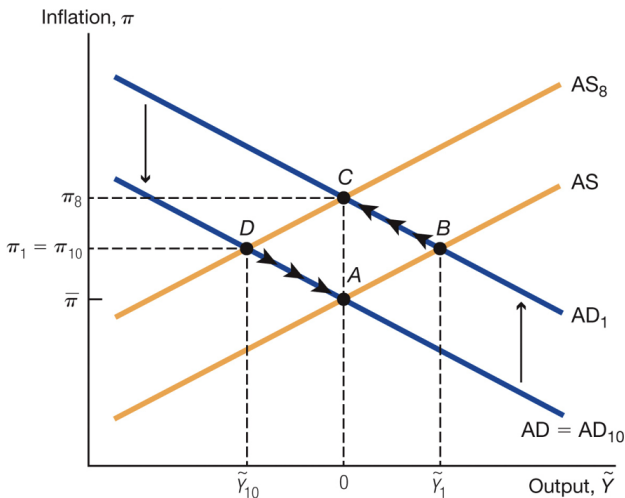
One Period Demand Shock



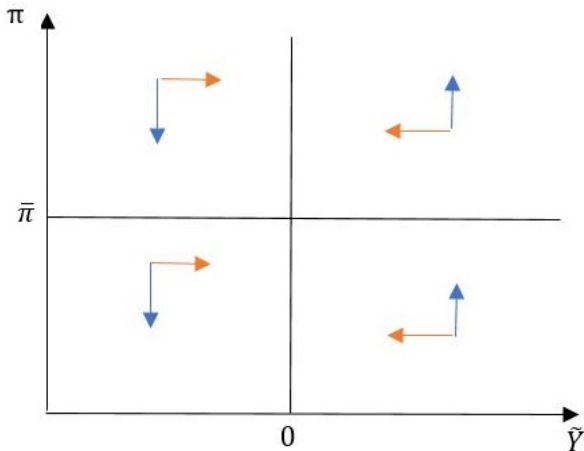
One Period Demand Shock



Multiple Period Demand Shock

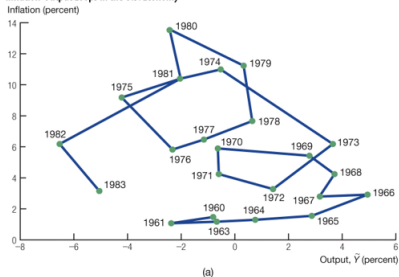


Phase diagram cost push and demand shocks

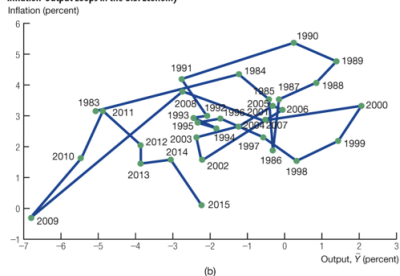


Empirical Evidence

Inflation-Output Loops in the U.S. Economy



Inflation-Output Loops in the U.S. Economy



Rules vs. Discretion

- Explicit Policy Rules:
 - ▶ (pros) Help anchor inflation expectations and maintain a stable inflation environment where shocks are deflected quickly
 - ▶ (cons) To work, policy rules must be credible, and, therefore, provide little flexibility to deviate from targets
- Discretionary Monetary Policy:
 - ▶ (pros) Provides flexibility to pursue expansionary economic policy
 - ▶ (cons) Shocks can be very disruptive as inflation pressures do not dissipate quickly
- It is extremely important that, in Modern Economies, CBs are independent from the government in power